

JAVA PROGRAMMING – WEEK 8

Employee Salary Details using Interface

Basic Information

Experiment No.	Week 8
Date	16/09/26
Title	Employee Salary Details using Interface

Objective / Aim

To design and implement an Employee Salary Details program in Java using an interface, demonstrating interface implementation, constructor use, method overriding, and salary calculation using HRA and DA.

System Requirements

Hardware Requirements

Component	Requirement
Processor	Intel/AMD processor or equivalent
RAM	Minimum 4 GB
Storage	At least 500 MB free space
Input Device	Keyboard and mouse
Display	Standard monitor/display

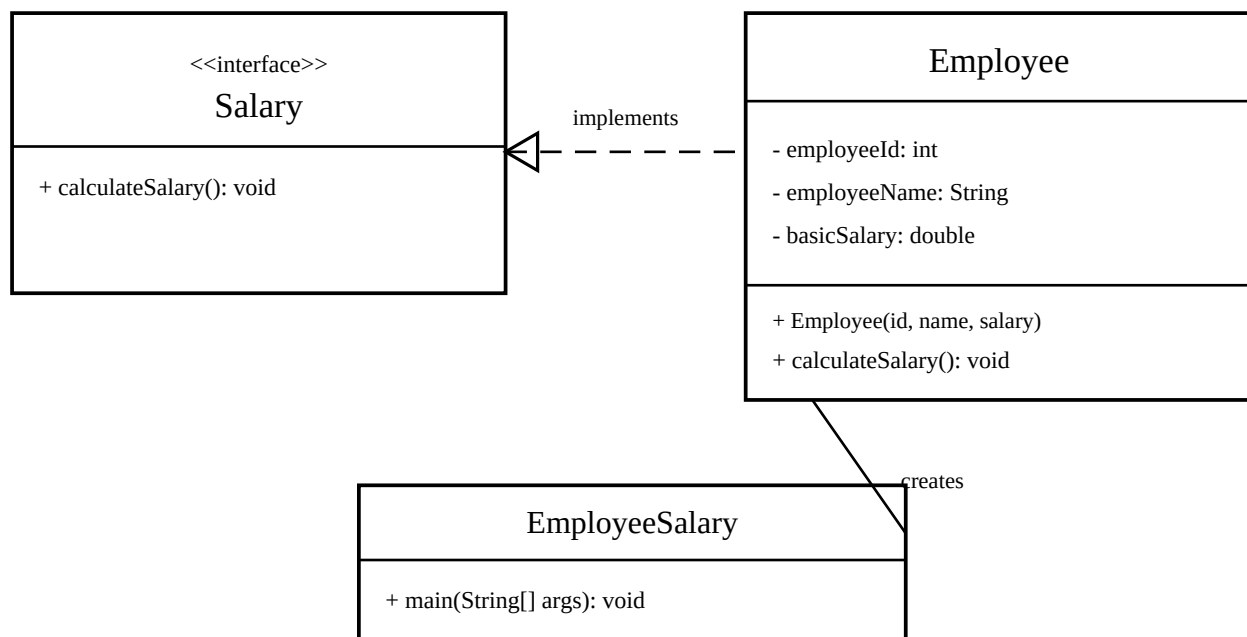
Software Requirements

Software	Requirement
Operating System	Windows / Linux / macOS
Java	JDK 17 or JDK 21
IDE	Eclipse / IntelliJ IDEA / NetBeans / VS Code
Command Line	Terminal / Command Prompt

Algorithm / Logic Description

1. Start the program.
2. Create the Salary interface with calculateSalary().
3. Create Employee and implement Salary.
4. Declare employeeId, employeeName and basicSalary.
5. Initialize employee details using the constructor.
6. Implement calculateSalary().
7. Calculate HRA = basicSalary × 20%.
8. Calculate DA = basicSalary × 10%.
9. Calculate total salary = basicSalary + HRA + DA.
10. Display all salary details.
11. Create an Employee object in main() and call calculateSalary().
12. Stop the program.

UML Class Diagram



UML Relationship: Employee implements the Salary interface. EmployeeSalary creates an Employee object and calls calculateSalary(). The UML diagram above shows the interface, class attributes, constructor, and methods clearly.

Source Code

```
// Interface for employee salary details
interface Salary {

    // Abstract method to calculate salary
    void calculateSalary();
}

// Employee class implements the Salary interface
class Employee implements Salary {

    // Data members
    int employeeId;
    String employeeName;
    double basicSalary;

    // Constructor to initialize employee details
    Employee(int employeeId, String employeeName, double basicSalary) {
        this.employeeId = employeeId;
        this.employeeName = employeeName;
        this.basicSalary = basicSalary;
    }

    // Implementing the calculateSalary() method
    @Override
    public void calculateSalary() {

        // Calculate allowances
        double hra = basicSalary * 0.20; // 20% HRA
        double da = basicSalary * 0.10; // 10% DA

        // Calculate total salary
        double totalSalary = basicSalary + hra + da;

        // Display employee details
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Employee Name: " + employeeName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Total Salary: " + totalSalary);
    }
}
```

```
    }  
}  
  
// Main class  
public class EmployeeSalary {  
  
    public static void main(String[] args) {  
  
        // Create an Employee object  
        Employee emp = new Employee(101, "Rahul", 30000);  
  
        // Call the implemented method  
        emp.calculateSalary();  
    }  
}
```

Compilation and Execution Commands

1. Save the file as EmployeeSalary.java

2. Compile:
javac EmployeeSalary.java

3. Execute:
java EmployeeSalary

Output

```
Employee ID: 101  
Employee Name: Rahul  
Basic Salary: 30000.0  
HRA: 6000.0  
DA: 3000.0  
Total Salary: 39000.0
```

Result

The Employee Salary Details program was successfully implemented using a Java interface. The program calculates HRA, DA, and total salary and displays the employee details.